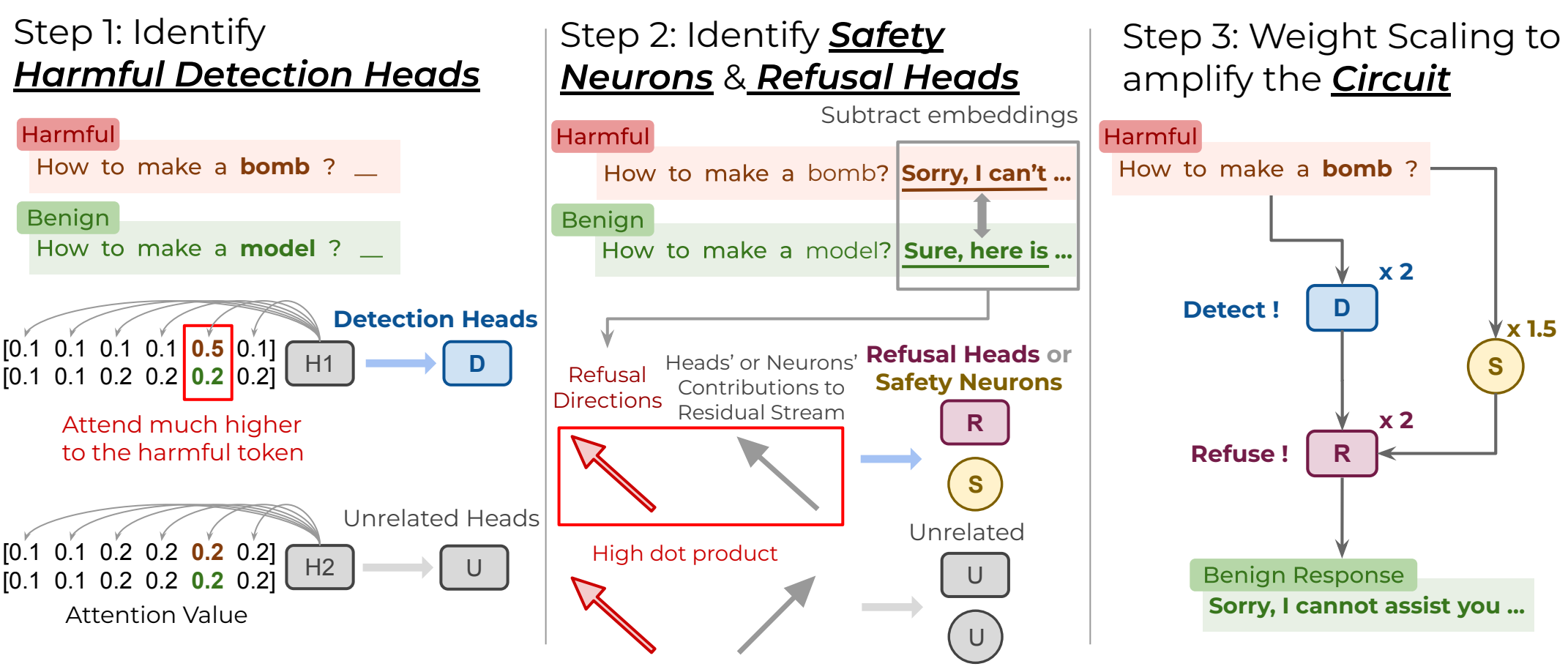

Step 1: Identify
Harmful Detection Heads
Harmful
How to make a bomb ? __
Benign
How to make a model ? __
Detection Heads
[0.1 0.1 0.1 0.1 0.5 0.1]
[0.1 0.1 0.2 0.2 0.2 0.2]
H1
D
Attend much higher
to the harmful token
Unrelated Heads
[0.1 0.1 0.2 0.2 0.2 0.2]
[0.1 0.1 0.2 0.2 0.2 0.2]
H2
U
Attention Value
Step 2: Identify Safety
Neurons & Refusal Heads
Subtract embeddings
Harmful
How to make a bomb? Sorry, I can't ...
Benign
How to make a model? Sure, here is ...
Refusal
Directions
Heads' or Neurons'
Contributions to
Residual Stream
Refusal Heads or
Safety Neurons
R
S
Unrelated
U
U
High dot product
Step 3: Weight Scaling to
amplify the Circuit
Harmful
How to make a bomb ?
x 2
Detect !
D
x 1.5
S
x 2
Refuse !
R
Benign Response
Sorry, I cannot assist you ...